Supplement: case selection, replay and audit trail

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Internal research draft from a living repository. Release and submission are Barry W. Brook's decision.

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S1. Case selection

The candidate pool is `papers/infrastructure-audit/evidence/claims.tsv`: 189 findings compiled read-only on 25 October 2026 from the programme's database, package and reference catalogues. The twelve cases were chosen on 8 November 2026 (`case-selection.md`) by five rules:

1. consequence for inference before defect status;
2. one case may hold several ledger rows;
3. coverage of the finding classes, including positive controls;
4. disclosure must be feasible;
5. findings of little consequence, and anything whose description could cause harm, are left out.

Four reserves were named in case a case failed replay: RevBayes's disabled fossilised birth–death range process, the optimal linear estimation interval's undercoverage in one package, a join error in a scraped diversity table, and a silent identity rotation in a reconstruction library. None was needed.

S2. Replay and independent check, by case

Each case was replayed on 8 November 2026 from pinned inputs. The replay records are in `papers/infrastructure-audit/evidence/replay-20261108/`, in four parts:

- `main/`: cases 1 and 11, and the parent checks;
- `data/`: cases 2, 3, 4 and 10;
- `software/`: cases 5, 6, 8 and 9;
- `calibration/`: cases 7 and 12.

Each record states:

- the input hashes;
- the command;
- the separate route by which a different actor rederived the result:
  - a fresh raw-data calculation;
  - a fresh pinned implementation;
  - reaggregation of retained outputs;
  - a mathematical or control check.

The `basis` column of the claims ledger names the route for each proposition, and the `limit` column states what it does not show.

What counts as independent. The original findings were made in one set of sessions. The replays and their checks were made by other actors using separate aggregation, interpolation, geometry or scoring code. Some of those actors use the same model family as the original auditors. The checks are therefore independent computational routes, not independent biological datasets, and they are not evidence that using two model families helps.

Software environment. R 4.6.1 with admtools 0.7.0, fossilbrush 1.0.6 and Rextinct 1.1.0; pyGPlates 1.0.0; content-pinned container images for the software runner. Full versions are recorded in each replay record.

S3. Statements withdrawn or narrowed during replay

Case	Original statement	Found by	Replaced by	Record
3	1,454 changed means, 423 by at least one degree; two per-bin class flips; 11–13% sample inflation	Codex replay, 8 Nov 2026	1,388 changed means, 411 by at least one degree; class flips as in C045	`evidence/replay-20261108/dat`
4	Sample count including a header row	Codex parent count	61,862 samples	`evidence/replay-20261108/mai`
5	"Wrong times above every hiatus"	Codex replay	Errors in affected interpolation segments only	C096 limit

Case	Original statement	Found by	Replaced by	Record
7	PyRate intervals “too narrow”	Codex replay	Undercoverage from location bias; widths not too narrow	C107 limit
10	5,291 damaged histories	Codex replay	2,967 altered histories; 5,291 is the count of cells in scientific notation	C035 limit
1	Records described as “withheld”	Codex replay	Absent from a dated public snapshot; cause not established	C191 limit

S4. Reproduction

From the repository root:

```
python3 papers/infrastructure-audit/tools/claims.py check \
  papers/infrastructure-audit/manuscript/claims.csv \
  papers/infrastructure-audit/manuscript/text/{main,supplement,brief,for-agents}.md
```

Each ledger row carries its own `reproduce` command and the commit at which its evidence was verified. Third-party archives are not redistributed; each replay record names the source and its hash, so a reader can obtain and verify the same input. Records absent from public database snapshots are reported as counts only and are not reproduced here.

References